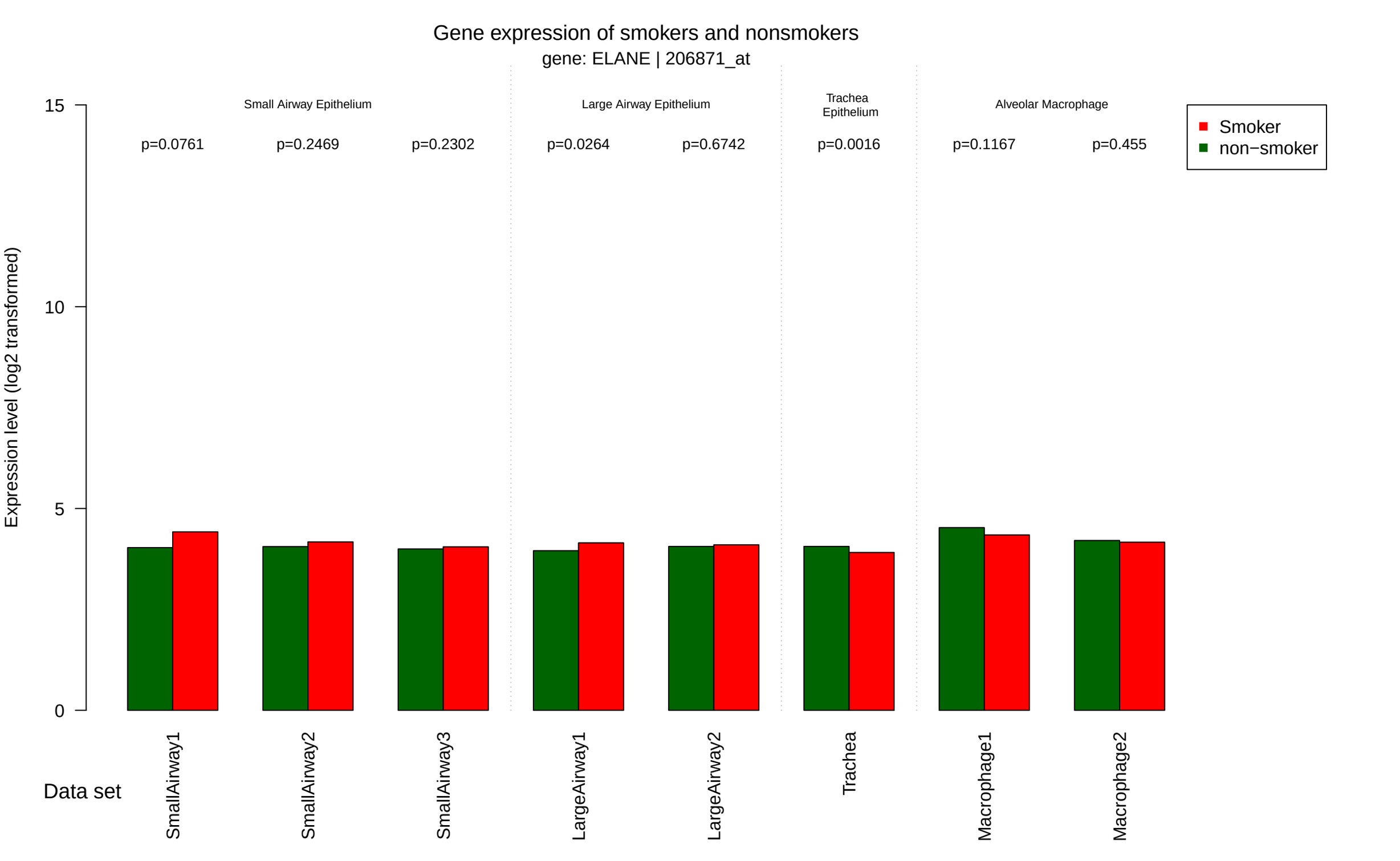
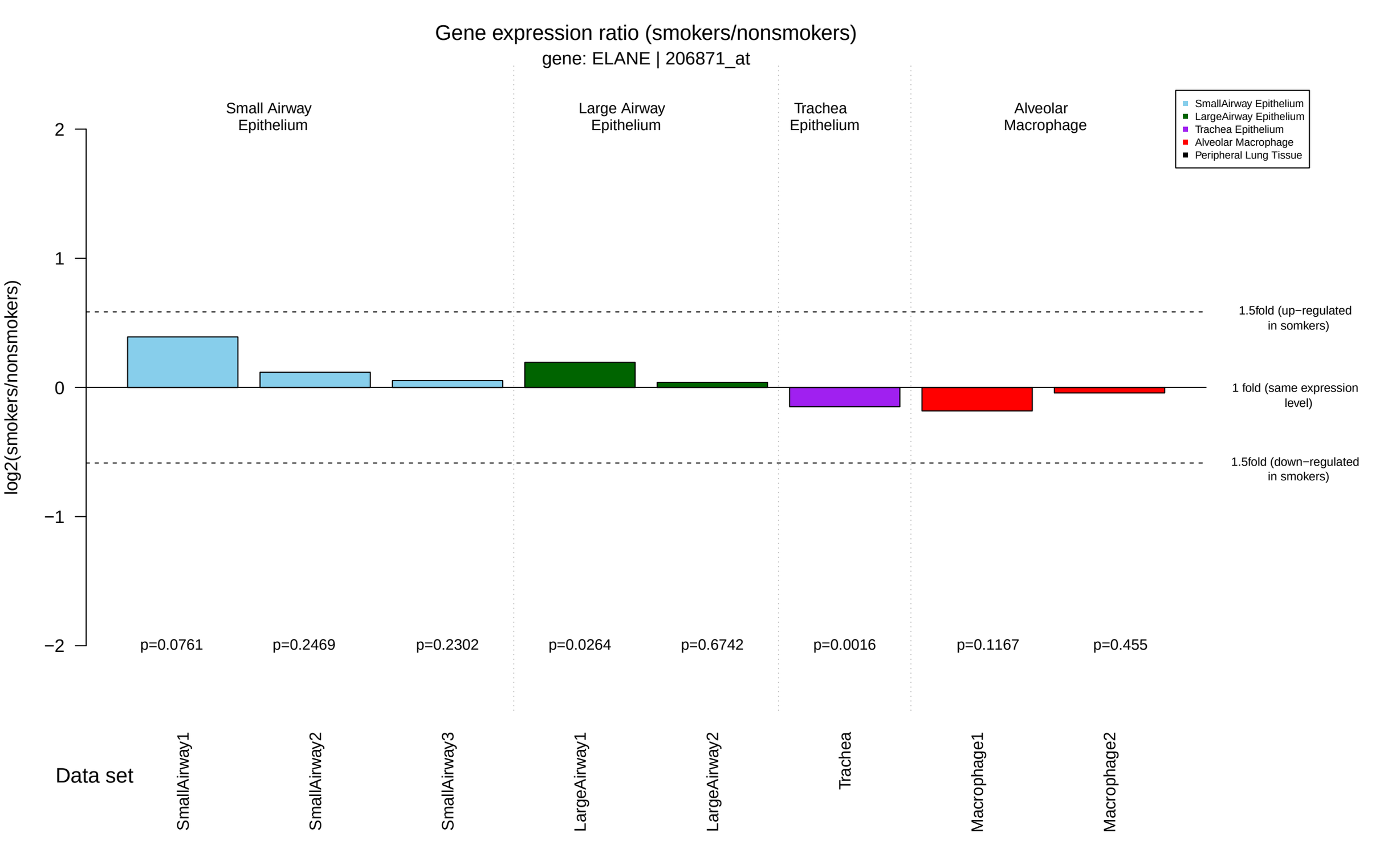
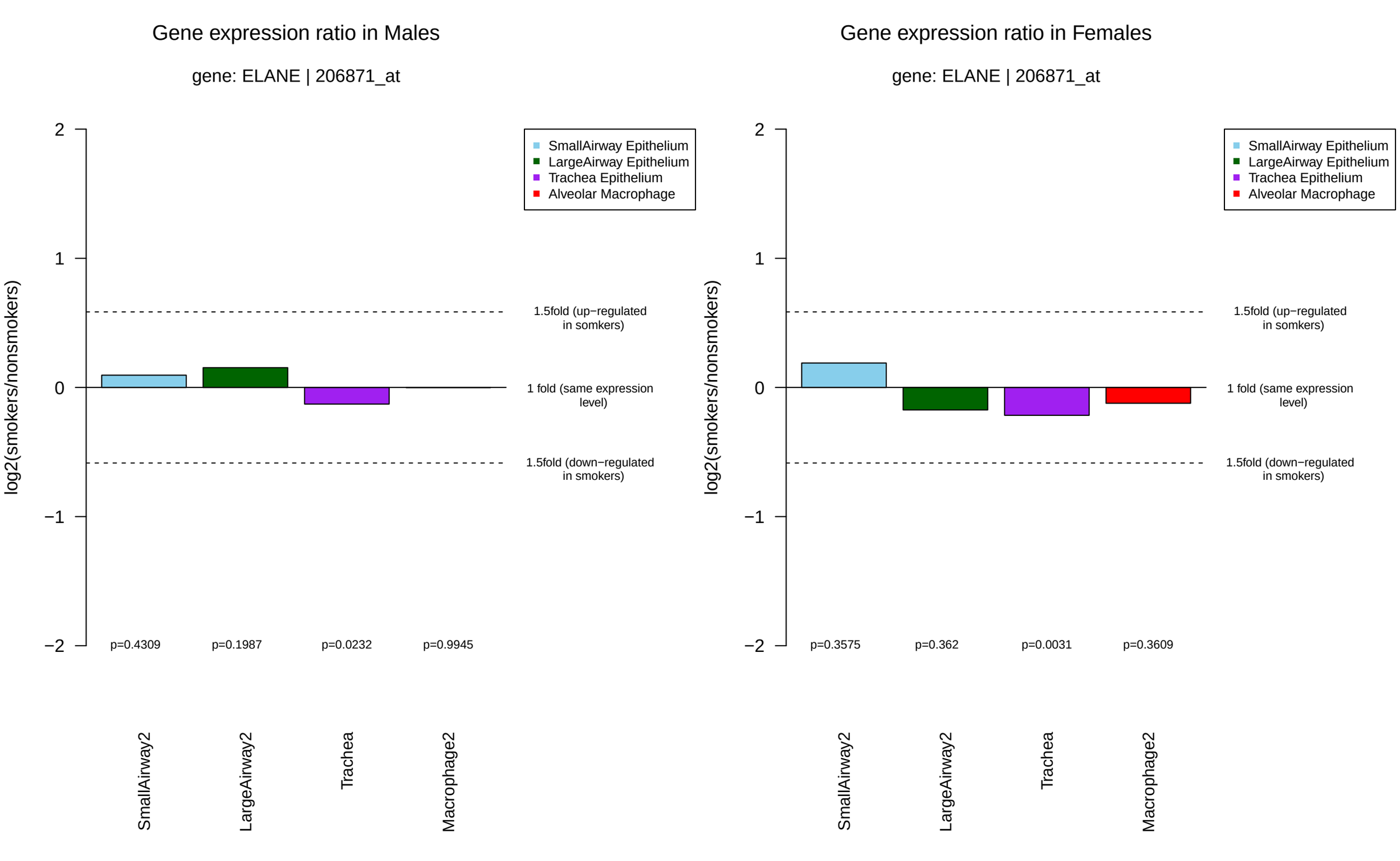
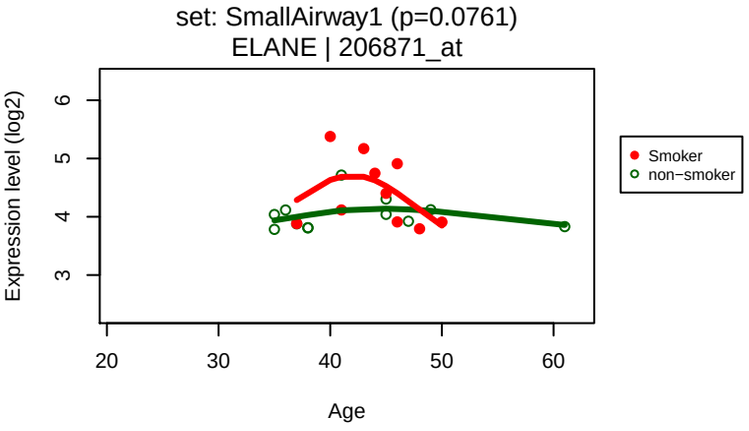
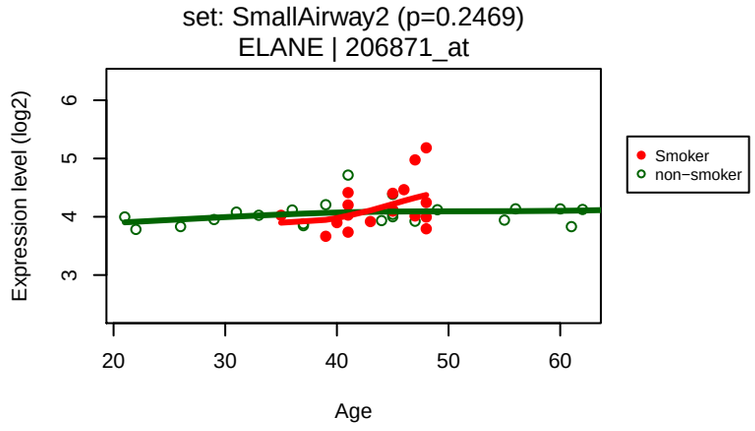
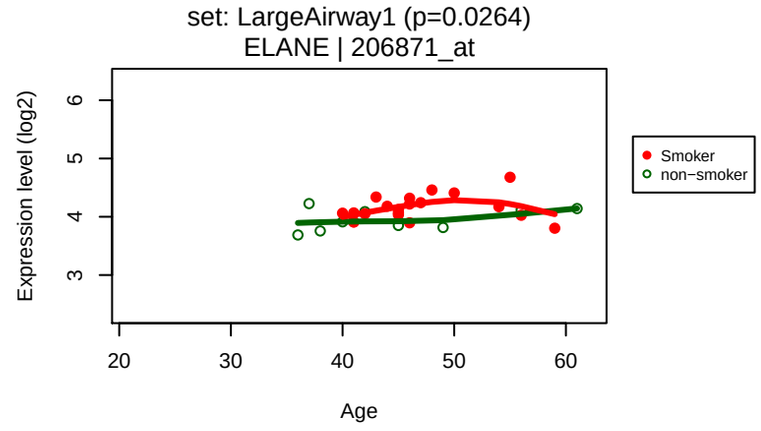

Gene expression of smokers and nonsmokers
gene: ELANE | 206871_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0761
p=0.2469
p=0.2302
p=0.0264
p=0.6742
p=0.0016
p=0.1167
p=0.455
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

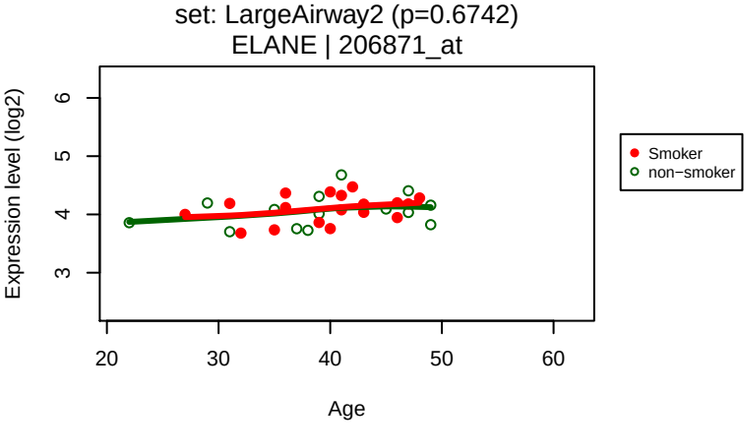

Gene expression ratio (smokers/nonsmokers)
gene: ELANE | 206871_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0761
p=0.2469
p=0.2302
p=0.0264
p=0.6742
p=0.0016
p=0.1167
p=0.455
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

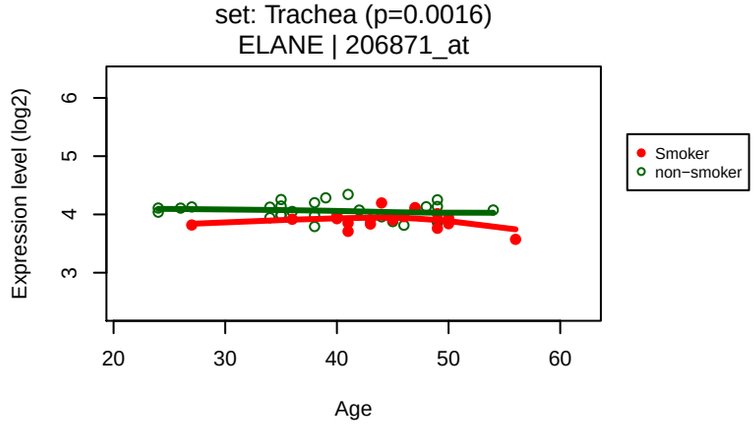

Gene expression ratio in Males
gene: ELANE | 206871_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4309
p=0.1987
p=0.0232
p=0.9945
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ELANE | 206871_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3575
p=0.362
p=0.0031
p=0.3609
SmallAirway2
LargeAirway2
Trachea
Macrophage2

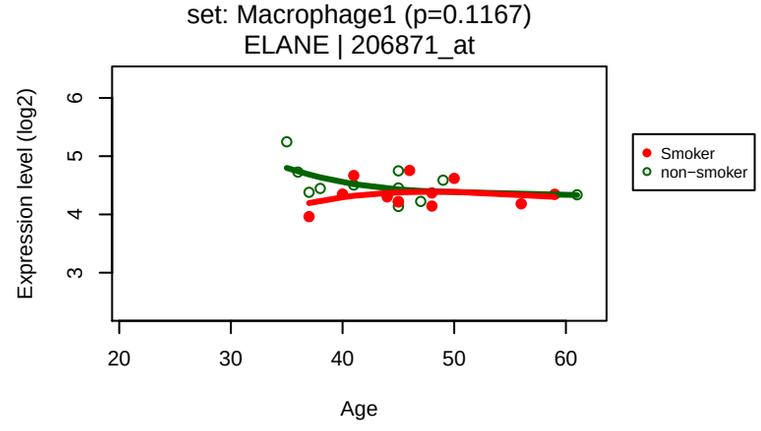

set: SmallAirway1 (p=0.0761)
ELANE | 206871_at
Expression level (log2)
Age
Smoker
non−smoker

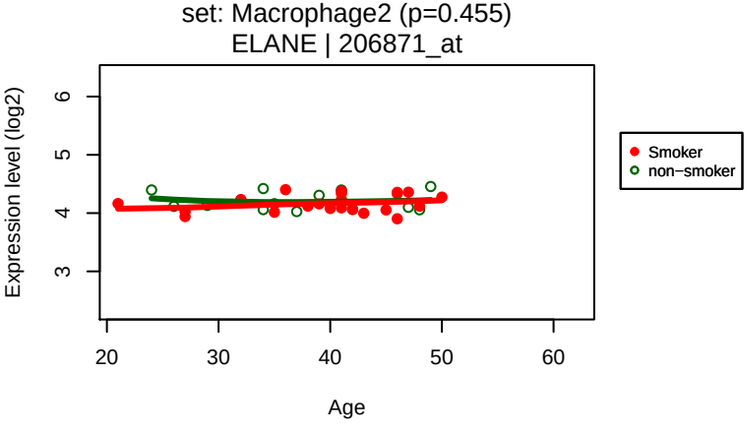

set: SmallAirway2 (p=0.2469)
ELANE | 206871_at
Expression level (log2)
Age
Smoker
non−smoker

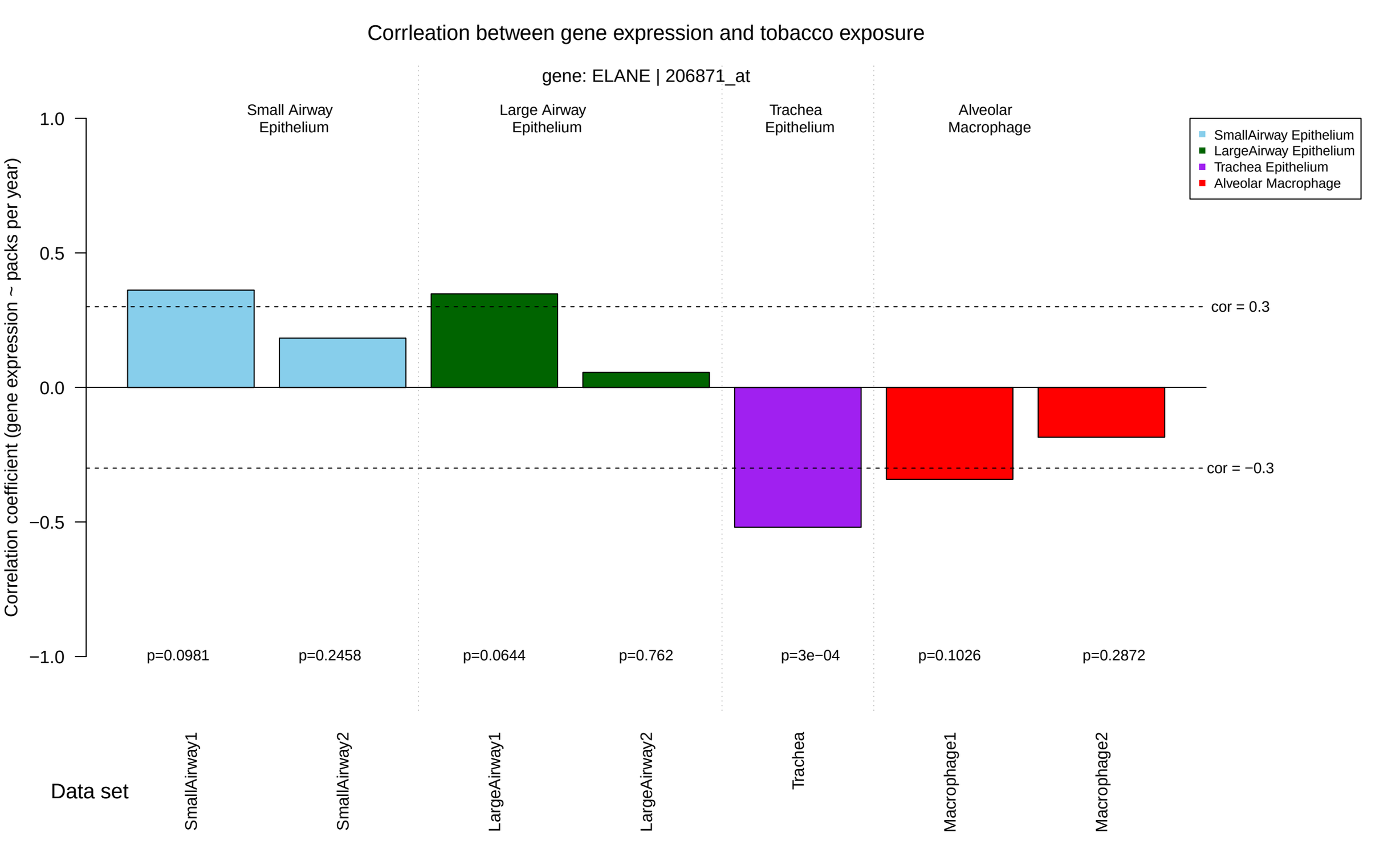

set: LargeAirway1 (p=0.0264)
ELANE | 206871_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6742)
ELANE | 206871_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0016)
ELANE | 206871_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1167)
ELANE | 206871_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.455)
ELANE | 206871_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ELANE | 206871_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0981
p=0.2458
p=0.0644
p=0.762
p=3e−04
p=0.1026
p=0.2872
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2